

Joint Illumination, Power, and Band Allocation for Multi-Beam LEO Satellites With Beam-Hopping Using Mixed-Integer Linear Programming

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Abstract—The paper presents a novel optimization approach for resource allocation in non-geostationary orbit (NGSO) satellite constellations employing beam-hopping (BH) payloads for large-scale broadband services. Current literature has proposed several optimization strategies to deal with the intricacies of BH applied to large satellite constellations. However, most works limit their study to only sub-portions of the complete problem, often fixing certain variables, such as frequency or power, or disregarding specific objectives, such as user fairness or excess of allocated resources. To address this gap, this work introduces a novel mixed-integer linear programming (MILP) formulation that integrates all available degrees of freedom, namely, power, frequency, and beam illumination, while optimizing for performance, quality of service and fairness through three objectives: unserved capacity (UC), extra-served capacity (EC), and time-to-serve (TTS). To reduce the computational complexity, an alternative time-split MILP is proposed by partitioning the problem into smaller time-based subproblems. The time-split MILP maintains the performance of the full MILP formulation in larger user scenarios and balances execution time with superior computational performance. Simulations demonstrate that the time-split MILP method surpasses other approaches, including deterministic (DB) and meta-heuristic (GA) solutions. It achieves up to a 6% reduction in UC, maintains EC levels below those

of DB and GA in low-user scenarios, and nearly halves the TTS, highlighting its efficiency in resource allocation and service optimization.

Index Terms—Satellite communications, non-terrestrial network, low earth orbit, resource allocation, beam-hopping, mixed-integer linear programming.

I. INTRODUCTION

IN RECENT years, satellite communication systems have experienced a transformative shift driven by emerging trends such as the deployment of small-sized satellites in low Earth orbit (LEO) and the advent of software-defined satellites with reconfigurable payloads [1]. These innovations, combined with more accessible commercial satellite systems, have enabled new services like direct satellite-to-handheld communication, satellite IoT, broadband access, and space edge computing. These capabilities align with 5G non-terrestrial networks (NTN) goals, leveraging satellite networks to enhance global connectivity [2], [3]. However, the dynamic and growing traffic demand necessitates advanced resource allocation strategies to optimize power, bandwidth, and coverage efficiently, especially critical in scenarios with limited onboard power and diverse quality-of-service (QoS) requirements [4], [5], [6], [7], [8]. In this context, beam-hopping (BH) payloads, which dynamically reallocate resources across service areas, emerge as a promising solution for non-geostationary orbit (NGSO) constellations, facilitating scalable and flexible broadband services.

BH enables the sequential illumination of specific areas (cells) with dynamically allocated power and bandwidth, allowing satellites to target high-demand regions as needed [9], [10]. Unlike traditional static beam configurations, BH enhances resource efficiency, reduces idle time, and optimizes load distribution, leading to lower latency and higher throughput. Despite these benefits, the operational complexity of BH arises from the need for real-time decisions on beam activation and resource allocation, balancing power constraints, varying traffic demands, and QoS diversity. Moreover, the temporal dimension of resource allocation adds an additional layer of complexity, requiring optimization not only of spatial coverage and resource distribution but also of time slots to adapt to fluctuating demand patterns effectively. The variety of possible performance metrics, such as maximizing throughput, minimizing underused resources, or ensuring fairness among users

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and beams, often involve trade-offs that need to be balanced for optimal performance.

The complexity of the BH resource allocation optimization problem in satellite networks has driven extensive research, leading to the development and application of diverse techniques for its resolution. Some authors have focused on studying the problem through metaheuristic optimization, using particle swarm optimization (PSO) or genetic algorithm (GA) approaches. For geostationary orbit (GEO) satellites, earlier works [11], [12] emphasized single-objective optimization, targeting capacity mismatch or unmet demand with basic GA crossover and mutation operators. Later studies introduced multi-objective frameworks to consider throughput and delay, while in [13] environmental dynamics like rain attenuation are incorporated. For LEO satellites, the focus shifted to addressing dynamic traffic distributions and user-centric metrics. In [14], multi-objective optimization is incorporated for throughput, fairness, and delay, integrating strategies like elite retention and interference clustering. Service metrics such as satisfaction and waiting time, are added in [15] adapting GAs for higher adaptability. The variety of methods reflects the differing system constraints and optimization goals, ranging from static to dynamic environments and from single to multi-objective formulations. However, these approaches share a critical limitation: they do not provide a quantifiable optimality gap, making it challenging to assess their true performance.

Several researchers have recently transitioned into machine learning (ML)-based approaches, such as deep reinforcement learning (DRL)-based algorithms. The application of DRL to the allocation of GEO satellite resources was pioneered in [16], focusing on dynamic beamforming to maximize system throughput under varying traffic conditions. This work is expanded in subsequent studies [17], [18], incorporating multi-agent DRL frameworks to improve scalability and coordination across multiple beams. In [19] the focus to LEO satellite constellations is shifted, employing DRL to tackle rapid topology changes and dynamic traffic distributions, introducing reward shaping techniques to enhance convergence speed. The integration of DRL with heuristic search to optimize resource allocation for hybrid GEO-LEO networks is introduced in [20], effectively balancing trade-offs between computational efficiency and solution quality. In [21] a DRL-based framework is proposed to address inter-satellite interference, achieving robust performance under uncertain environmental conditions. Still in LEO, the authors in [22] present a two-tier DRL-based optimization problem, optimizing BH and power allocation to maximize throughput and minimize inter-cell service delay. DRL-based methods represent a shift toward adaptive learning-based resource-allocation strategies for BH systems and have demonstrated strong performance in dynamic settings. However, similar to other metaheuristic approaches, they do not guarantee optimality. The problem formulations of the analyzed works are tailored to specific decision variables or performance targets that differ from the joint BH resource allocation problem considered in this work.

Accordingly, others have focused on solving the beam-hopping problem through analytical exact optimization

approaches, ensuring optimality at the expense of increasing computation time. In [23] a mixed integer linear programming (MILP) formulation is presented at GEO level, optimizing the BH illumination sequence and the assignment of band resources within each cluster. The power allocation is assumed to be constant for each beam, a particularity that can not be translated to NGSO low-power cubesat platforms. Within exact optimization methodologies, a binary quadratic programming (BQP) framework has also been developed for GEO satellites by [24], where the illumination pattern is optimized through a clustered precoded beam scheme, to minimize the interference-based penalty and to satisfy a certain beam demand in a given time window. The authors in [25] introduce a mixed-integer conic programming (MICP) problem, to minimize the gap between the served and requested capacity utilizing BH and non-orthogonal multiple access (NOMA) techniques at LEO level. The formulation addresses power allocation, beam scheduling, and terminal time slot assignment. In [26] a MILP beam scheduling and power allocation problem is presented, for LEO, minimizing the throughput for a fixed bandwidth allocation. In [27], the authors consider a LEO BH system where beam-hopping scheduling, frequency-band assignment, and transmit-power allocation are jointly optimized. The problem is reformulated as an MILP and solved using a Tree Markov Decision Process (TMDP)-based algorithm. The objective, however, is to maximize the overall user rate under covertness constraints imposed by an eavesdropping GEO satellite and fixed minimum-rate requirements. Service-availability and timing-related metrics (such as TTS), as well as explicit control of unserved and extra-served capacity (UC/EC), are not incorporated. The approach is applied to relatively low-dimensional instances (with few beams, users, bands, power levels, and time slots), as its scalability to larger LEO BH scenarios, such as those considered in this paper, remains intractable. Their approach relies on an enumeration-based linearization, which, under the dimensions considered in this work, would require pre-computing on the order of 1.7×10^{181} (reduced) and 2.5×10^{596} (enlarged) feasible joint configurations before the MILP can be constructed.

Overall, these works limit their study to only sub-portions of the complete BH optimization problem, often fixing variables such as power, frequency, or illumination, or omitting objectives related to user fairness or excess power allocation. As summarized in Table I, the existing literature spans a diverse set of formulations and resolution methodologies, highlighting the need for a multidimensional approach that achieves optimality while remaining scalable to the dynamic conditions of LEO satellite networks. This paper introduces an exact analytical optimization framework for the BH problem using MILP, jointly optimizing illumination sequence, bandwidth, and power allocation under constraints from antenna architecture, interference, and limited power and spectrum resources. The optimization method encodes variables, constraints and decisions using a novel formulation based on a weighted cost function to minimize unserved capacity and response time while avoiding excess resource allocation beyond the requested capacity—an important metric not considered in pre-

TABLE I
SUMMARY OF STATE-OF-THE-ART APPROACHES FOR BEAM-HOPPING OPTIMIZATION IN SATELLITE SYSTEMS

Reference	Strategy	Scenario	Decision Variables	Interference	Objective	Optimality	Scalability
[11]–[13]	GA / PSO (SOEA)	GEO	III + BW or III + P	-	UC	-	Heuristic
[14], [15], [28]	GA (MOEA)	LEO/GEO	III + BW or III + P	Implicit: adjacency/clustering	UC + TTS	-	GA pruning / clustering
[16], [17]	DRL	GEO	III	Implicit: reward shaping	UC + TTS	-	Model-free DRL
[18]–[22]	DRL	GEO/LEO	III + BW or III + P	Explicit: inter-beam / overlap	UC + TTS	-	Multi-agent / hybrid DRL
[23]	MILP	GEO	III + BW	Explicit: dual-pol coupling	UC	Global	MILP solver
[24]	BQP/SDP	GEO	III	Implicit: precoding/penalty	UC	SDP rank-1	SDP relax + greedy
[25]	MICP	GEO	III + P	Explicit: NOMA user interference	UC + EC	Bounds	Bounding + heuristics
[26]	MILP	LEO	III + P	-	UC + TTS	Nash Equilibrium + local	Potential game + interior-point
[27]	MILP	LEO	III + BW + P	Explicit: intra-/inter-user interference	UC	-	Limited, enumeration-based linearization
This work	MILP	LEO	III + BW + P	Implicit: inter-beam adjacency avoidance	UC + EC + TTS	Optimal (full) close-to-optimal (split)	Time-split MILP for large dimensional problems

vious works, that is crucial for constrained satellite payloads, which follows the antenna model proposed by the authors in [10]. Proper normalization and weighting factors in the objective function are carefully motivated and presented.

While the MILP formulation delivers exact solutions in low-dimensional settings, therefore allowing for performance benchmarking, its computational complexity grows rapidly with the dimensionality of the problem. For high-dimensional, and therefore computationally complex scenarios, a novel time-split MILP resolution is introduced, bounding the time horizon into disjoint subspaces and solving the analytical and therefore exact BH optimization problem independently over time. Crucially, it aligns more naturally with the dynamic nature of LEO satellite networks: as satellites move and new users and service cells come into scope, the optimization problem evolves. The time-split method enables cumulative decision-making across time, effectively capturing these time-varying conditions, something that full-horizon static optimization fails to do.

Despite its reduced complexity, the time-split MILP provides close-to-optimal solutions compared to the developed full MILP formulation. These methodologies highlight the trade-offs between optimality and computational feasibility, addressing scenarios of varying complexity.

The MILP framework provides a reliable reference for assessing the effectiveness of beam-hopping optimization methods in satellite resource allocation, including heuristic and ML-based strategies. Importantly, it enables the quantification of optimality gaps, providing essential insights into how far individual solutions are from the theoretical optimum. This capability not only enhances the assessment of current methods, but also lays the groundwork for the development of data-driven approaches, such as DRL, by offering a reliable

reference to guide training toward more efficient and reliable resource allocation solutions. To the best of the authors' knowledge, no other MILP-based approaches in the literature address such a flexible problem formulation.

The main contributions of the paper are:

- A joint MILP formulation that simultaneously optimizes illumination scheduling, bandwidth allocation, and power distribution in beam-hopping systems, with implicit adjacency-based interference avoidance as well as hardware and energy constraints.
- The introduction of extra-served capacity (EC) as a minimization metric, treated as a first-class objective alongside unserved capacity (UC) and time-to-serve (TTS). This enables the explicit analysis of trade-offs between service capacity and energy efficiency when the resources allocated in excess of user demand are controlled.
- A time-split MILP variant that preserves per-window optimality while bounding computational complexity, thus allowing practical scalability to larger problem dimensions.
- An extensive performance analysis across different scenarios, evaluating the movements of the three key performance indicators (KPIs): UC, EC, TTS and including additional resolution strategies, such as deterministic demand-based (DB) and meta-heuristic, genetic algorithm (GA) based solutions developed by the authors, where convergence speed and solution optimality is balanced.

The remainder of the paper is structured as follows. Section II describes the system model, input parameters, and assumptions. Section III presents the MILP formulation, introducing decision variables, constraints, and the objective

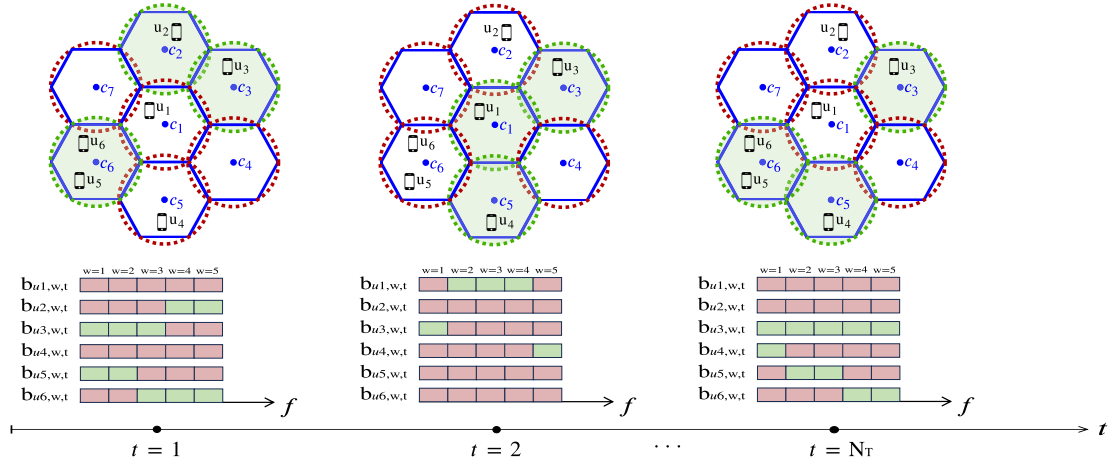


Fig. 1. Beam-hopping illumination scheme. Example for $N_{\text{ill}} = 3$, $N = 5$, $N_C = 7$, $N_U = 6$ and N_T .

function. This section also discusses the alternative time-split MILP and its impact on computational complexity. Section IV analyzes two scenarios, assessing the impact of weighting parameters and convergence on performance, and compares DB, GA, time-split MILP, and full MILP in both low- and high-dimensional cases. Finally, Section V concludes the paper.

II. SYSTEM MODEL

A satellite under NGSO orbit is considered. Within its field of view, the satellite illuminates a portion of the ground through the use of spot beams, a portion that is referred to as a cell. Cells represent the geographic area covered by a single radio beam and their distribution is dependent on the total cell distribution set, $\mathcal{C}$. In this work, an Earth-Fixed Cell (EFC) layout is adopted, consistent with 3GPP NTN architectures [29], where the number of rings and cells (i.e., the resolution) can be adjusted over the satellite FoV.

The user set $\mathcal{U}$, containing a total of N_U users, is randomly generated once per scenario within $\mathcal{C}$. The requested capacity associated to user u , at time instant t is characterized by $R(u, t)$, with $u \in \mathcal{U}$ and $t \in \mathcal{T}$. Users are aggregated into cells, as the illumination granularity is fixed at cell level. For a given snapshot, the user per cell matrix, $\mathbf{UpC}$, characterizes the relationship between users, with $\text{UpC}(u, c) = 1$ if user $u \in \mathcal{U}$ is aggregated to cell $c \in \mathcal{C}$ and $\text{UpC}(u, c) = 0$ otherwise.

The satellite can only illuminate a sub-portion of beams at a time. Through the BH technique, the total simulation time, T_T , is divided into N_T time slots. As LEO satellites move with respect to a fixed observer point on Earth's surface, to avoid handovers, T_T is determined for the duration where an invariant snapshot can be considered. In total, there are N_T time slots within the time slot set $\mathcal{T} = \{t_1, t_2, \dots, t_{N_T}\}$. As depicted in Fig. 1, N_{ill} cells are sequentially illuminated from the total N_C set of cells, a value directly linked with the on board satellite communication architecture [9]. As it will be seen, the set of illumination schemes is optimized for all the sequences of time slots, evaluating overall traffic and time-based performance indicators.

The beam-hopping-based resource allocation problem is studied for the user downlink transmission. The satellite

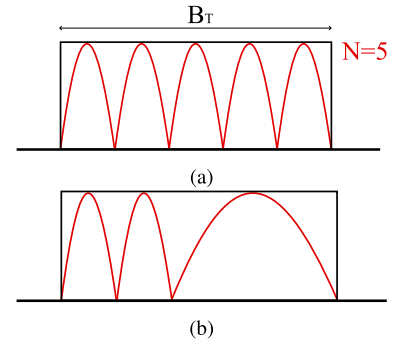


Fig. 2. Total bandwidth (B_T) and $N = 5$ frequency bins: (a) Frequency bin per user. (b) Two single frequency bins for two different users and aggregation of three frequency beams to a given user.

operates over a spectrum divided into frequency channels. The total bandwidth B_T is partitioned into N equal frequency bins, indexed by the set $\mathcal{N} = \{1, 2, \dots, N\}$, as illustrated in Fig. 2. For each time slot, different subsets of $\mathcal{N}$ can be allocated to the users located within an illuminated cell. The simultaneous illumination of adjacent cells using the same frequency bin is restricted, ensuring that users within neighboring cells do not operate on identical frequencies at the same time. This is also represented in Fig. 1.

This adjacency restriction is effective in mitigating co-channel interference. In the worst-case configuration—two adjacent beams illuminated with a cell between—the carrier-to-interference ratio (C/I) obtained with the 3GPP antenna reference pattern [30] is around 17dB for the scenario presented. Since typical link budgets operate with carrier-to-noise ratio (C/N) values well below this C/I level, interference does not impose a practical restriction, and the adjacency rule is sufficient to ensure co-channel isolation. The C/I level can be further increased by well-known amplitude tapering techniques in the synthesis of the array pattern onboard, confirming that adjacency constraints primarily act as a combinatorial limitation on feasible allocations rather than a physical interference bottleneck.

As discussed in [10], when the number of cells (N_C) within the satellite FoV is large and beam-hopping induces temporal sparsity, the feasible solution space enlarges, since there are more possible illumination patterns (N_{ill}) that avoid

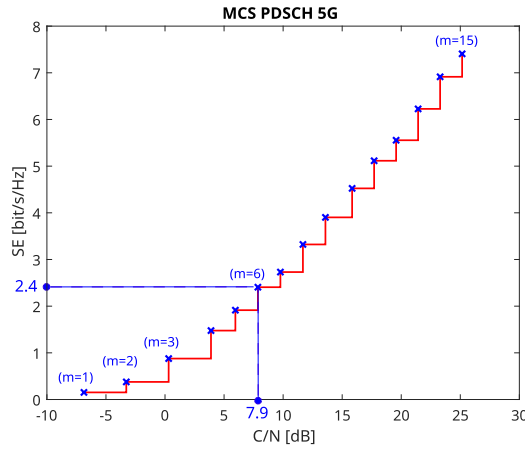


Fig. 3. Modulation coding schemes SE vs C/N.

adjacency conflicts under uniform or spread demand. In such cases, adjacency restrictions impose only a weak limitation on resource allocation. This effect does not hold in low- N_C schemes, where adjacency conflicts remain more constraining.

Based on the particularities of each scenario under study, the cell adjacency matrix is pre-computed, $\mathbf{AdjC}$. If two cells $c, q \in \mathcal{C}$ are considered adjacent to each other, then $\mathbf{AdjC}(c, q) = 1$. From cell adjacency, based on the performed user to cell aggregation, the user adjacency matrix is computed: $\mathbf{AdjU} = \mathbf{UpC}^T \mathbf{AdjC} \mathbf{UpC}$. The user adjacency matrix is a binary matrix encoding co-cell users. Formally, if user $u \in \mathcal{U}$ is located in cell $c \in \mathcal{C}$ and user $v \in \mathcal{U}$ is located in cell $q \in \mathcal{C}$, and $\mathbf{AdjC}(c, q) = 1$, then $\mathbf{AdjU}(u, v) = 1$. As it will be seen, this matrix will be essential for frequency bin assignment to restrict co-cell interference terms.

The satellite has the capability of adapting the modulation and coding scheme (MCS) to each of the beams as required. Fig. 3 shows the C/N requirements of the MCS formats defined in the physical layer of 5G [31]. The number of available MCSs is discrete and denoted by M , with the individual schemes indexed by the set $\mathcal{M} = \{1, 2, \dots, M\}$. Each MCS $m \in \mathcal{M}$ provides a different spectral efficiency (SE). In order to close the link budget, i.e., to ensure the required C/N, each $m \in \mathcal{M}$ has an associated minimum required power, which depends on the assigned bandwidth, the distance from user to satellite, and the user equipment. Thus, for a user-to-satellite link, an increase in power does not yield a higher spectral efficiency (SE) unless the modulation and coding scheme (MCS) changes. Consequently, the power levels of interest are discrete and correspond to the thresholds associated with each $m \in \mathcal{M}$. The required power value $P(u, m, n)$ for user $u \in \mathcal{U}$, under an assignment of $n \in \mathcal{N}$ frequency bins and a target MCS $m \in \mathcal{M}$, is directly derived from the link budget in (1).

$$\begin{aligned} C/N|_{\text{dB}}(u, m, n) &= G_{\text{TX}}(u) + P(u, m, n) - FSL(u) \\ &\quad + G_{\text{RX}}(u) - (K + T_{\text{eq,noise}}(u) + n \cdot (B_T/N)) \\ &\quad \forall u \in \mathcal{U}, \forall m \in \mathcal{M}, \forall n \in \mathcal{N} \end{aligned} \quad (1)$$

$G_{\text{TX}}(u)$ and $G_{\text{RX}}(u)$ represent the on board transmit antenna gain and user equipment receiving antenna gain, respectively. $FSL(u)$ term refers to free space losses, K to the Boltzmann

constant and $T_{\text{eq,noise}}(u)$ to the equivalent noise temperature of the user terminal.

As mentioned, the minimum required $P(u, m, n)$ by each modulation scheme, m , under a given frequency bin allocation, n is determined by the m MCS's C/N requirement: $C/N|_{\text{dB}} \geq C/N|_{\text{requirement}}(m)$. Once the link budget is closed by ensuring at least $C/N|_{\text{requirement}}(m)$, it is the associated m MCS's SE the one that fixes the offered capacity: $D(m, n)$. As it can be seen in Fig. 3, if 7.9dB of C/N is achieved, the 6th MCS can be selected, corresponding to a 16QPSK modulation format with a coding rate of 616/2024 as per in [31] and thus an associated SE of 2.4 bit/s/Hz. The odd MCS indices are taken from PDSCH Table II in [31].

In order to compute $D(m, n)$, the achievable symbol rate $R_s(n)$ must first be determined, which depends on the number of assigned frequency bins n out of the total bandwidth B_T . Equation (2) shows the translation from occupied bandwidth to symbol rate, where α denotes the roll-off factor of the baseband shaping filters.

$$R_s(n) = \frac{n \cdot (B_T/N)}{(1 + \alpha)}, \quad \forall n \in \mathcal{N} \quad (2)$$

The instantaneous offered capacity $D(m, n)$ in (3) then depends on the selected MCS m (through its spectral efficiency $SE(m)$), as depicted in Fig. 3) and the number of assigned frequency bins n , which determines $R_s(n)$ in (2).

$$D(m, n) = SE(m) \cdot R_s(n) \cdot (T_T/N_T), \quad \forall m \in \mathcal{M}, \forall n \in \mathcal{N} \quad (3)$$

Once these variables are pre-computed as a function of the distinct operation points (m, n) for each user u and time instant t , the optimization problem will later determine which of them to activate to comply with each of the objectives. The input variables introduced in this section are summarized in Table II, while the complete problem formulation is presented in the following section.

III. PROBLEM FORMULATION

The MILP formulation for the beam-hopping optimization problem is outlined in this section, introducing decision variables, constraints, and the objective function. Additionally, an alternative time-split MILP approach is introduced, with a discussion on its impact on computational complexity and the trade-offs between computational efficiency and solution quality.

The proposed MILP formulation addresses the beam-hopping optimization problem by determining the cell-based illumination sequence and, for each illuminated user, the allocation of frequency bins and the appropriate MCS scheme. The optimization objectives are threefold: (i) to maximize the served capacity, (ii) to minimize excess traffic above the requested capacity (extra-served capacity), and (iii) to reduce user waiting time by minimizing non-served time windows. To achieve this, both continuous and discrete decision variables are introduced, followed by the corresponding linearized constraints, and culminating in the definition of a multi-KPI objective function that balances traffic efficiency and temporal fairness.

TABLE II
INPUT SYSTEM VARIABLE NOMENCLATURE

Variable	Symbol
Number of simultaneous illumination beams:	N_{ill}
Number of users:	N_{U}
Number of cells:	N_{C}
Total simulation time:	T_{T}
Number of illumination time slots:	N_{T}
Total RF transmit power:	P_{T}
Total available MCSs:	M
Total BW:	B_{T}
Total available frequency bins:	N
User u in cell c :	$UpC(u, c)$
Adjacency between cell c and cell q :	$AdjC(c, q)$
Adjacency between user u and user v :	$AdjU(u, v)$
Requested capacity by user u , at time instant t :	$R(u, t)$
Offered capacity when selecting MCS m , and n frequency bins:	$D(m, n)$
Required power by user u , for MCS m , and n frequency bins:	$P(u, m, n)$

TABLE III
DESIGN VARIABLE NOMENCLATURE

Variable	Symbol	Type	Size
Illumination:	$i_{c,t}$	B	$N_{\text{C}} \cdot N_{\text{T}}$
MCS selection:	$m_{u,m,t}$	B	$N_{\text{U}} \cdot M \cdot N_{\text{T}}$
Band selection:	$b_{u,w,t}$	B	$N_{\text{U}} \cdot N \cdot N_{\text{T}}$
Operating point selection:	$z_{u,m,n,t}$	B	$N_{\text{U}} \cdot M \cdot N \cdot N_{\text{T}}$
Number of frequency bins:	$x_{u,n,t}$	B	$N_{\text{U}} \cdot N \cdot N_{\text{T}}$
Above the number of frequency bins:	$a_{u,n,t}^+$	B	$N_{\text{U}} \cdot N \cdot N_{\text{T}}$
Below the number of frequency bins:	$a_{u,n,t}^-$	B	$N_{\text{U}} \cdot N \cdot N_{\text{T}}$
Unserved capacity (UC):	$\gamma_{u,t}$	S	$N_{\text{U}} \cdot N_{\text{T}}$
Extra-served capacity (EC):	$f_{u,t}$	S	$N_{\text{U}} \cdot N_{\text{T}}$
Time-to-serve (TTS) counter:	$c_{c,t}$	I	$N_{\text{C}} \cdot N_{\text{T}}$

The nomenclature used in the optimization problem formulation is summarized in Table III. Inherent and auxiliary variables are summarized. The binary variables are represented as B type variables. I represents an integer variable. and S a semi-continuous variable. Finally, the dimensions of each variable are also depicted.

A. Simultaneous Illumination

As discussed in the previous section, the BH allows the operator to allocate power and bandwidth differently at each time-step, based on the set of cells illuminated. As such, one of the key decisions is the illumination scheme for each instance of time $t \in \mathcal{T}$. The illumination variable $i_{c,t}$ represents a boolean variable, activated only if the cell c is illuminated at time instant t . The number of cells that can be illuminated simultaneously is directly related to the radio-frequency (RF) architecture [9] on board the satellite, restricted to $N_{\text{ill}} = 4$ in the example of Fig. 1.

$$\sum_c i_{c,t} \leq N_{\text{ill}}, \quad \forall t \in \mathcal{T} \quad (4)$$

B. Modulation and Coding Format Selection

The BH algorithm must determine the MCS assigned to each illuminated user. The point in Fig. 3 the user $u \in \mathcal{U}$ will operate at time instant $t \in \mathcal{T}$ from the total $|\mathcal{M}| = 15$ available MCS formats. Once again, $m_{u,m,t}$ represents a boolean variable that must only be activated for the selected MCS format $m \in \mathcal{M}$. If the cell where the user u has been

aggregated is not illuminated at a given time instant t , no MCS must be assigned to that user and therefore: $m_{u,m,t} = 0, \forall m \in \mathcal{M}$.

$$\sum_m m_{u,m,t} \leq \sum_c UpC(u, c) \cdot i_{c,t}, \quad \forall t \in \mathcal{T}, \forall u \in \mathcal{U} \quad (5)$$

C. Frequency Band Assignment

The decision lies on deciding how many frequency bins and which ones are assigned to each illuminated user out of the total bandwidth B_{T} . Two new variables are introduced for these two purposes, the number of frequency bins variable, $x_{u,n,t}$, and the band selection variable, $b_{u,w,t}$, respectively. As the band is split in N bins, $b_{u,w,t}$ will take the value of 1 if subband $w \in \mathcal{N}$ is selected by user u at time instant t . The auxiliary variable $x_{u,n,t}$ is directly derived based on the number of 1s, or in other words, the number of frequency bins assigned:

$$x_{u,n,t} = \begin{cases} 1 & \text{if } \sum_w b_{u,w,t} = n \\ 0 & \text{otherwise} \end{cases}, \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (6)$$

To linearize the previous equation, two new binary activation variables are introduced, indicating whether the number of frequency bins variable is above or below its value, $a_{u,n,t}^+$, and $a_{u,n,t}^-$ respectively. The possible n values above and below the sum of the assigned frequency bins must be restricted, and as these two possibilities coexist, $a_{u,n,t}^+$ and $a_{u,n,t}^-$ are respectively required, $\forall u \in \mathcal{U}, \forall n \in \mathcal{N}$ and $\forall t \in \mathcal{T}$. Note that these two parameters are just a mathematical mechanism to represent an integer value as a collection of binary variables where only one is active at a time and hold no physical meaning.

$$a_{u,n,t}^+ = \begin{cases} 1 & \text{if } \sum_w b_{u,w,t} > n \\ 0 & \text{otherwise} \end{cases}, \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (7)$$

$$a_{u,n,t}^- = \begin{cases} 1 & \text{if } \sum_w b_{u,w,t} < n \\ 0 & \text{otherwise} \end{cases}, \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (8)$$

These two variables are activated depending on the case. If $(\sum_w b_{u,w,t} - n)$ is positive, we are in the above case and therefore $a_{u,n,t}^+$ is required to be activated, according to (9). Similarly, if the above expression is negative, then, as per (11), $a_{u,n,t}^-$ is activated. As (9) does not force $a_{u,n,t}^+$ to be 0 when $(\sum_w b_{u,w,t} - n) \leq 0$, the lower pair of (10) is imposed. Similarly, through (12), when $(\sum_w b_{u,w,t} - n) \geq 0$, $a_{u,n,t}^-$ is deactivated.

$$\sum_w b_{u,w,t} - n \leq N \cdot a_{u,n,t}^+, \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (9)$$

$$\sum_w b_{u,w,t} - n \geq 1 - N \cdot (1 - a_{u,n,t}^+), \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (10)$$

$$\sum_w b_{u,w,t} - n \geq -N \cdot (a_{u,n,t}^-), \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (11)$$

$$\sum_w b_{u,w,t} - n \leq N \cdot (1 - a_{u,n,t}^-) - 1, \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (12)$$

Finally, the consistency between $b_{u,w,t}$ and $x_{u,n,t}$ is ensured by using the pair of activation variables. According to (6), $x_{u,n,t}$ must only be activated for the exact value of $n \in \mathcal{N}$.

By following the logic explained above, if $a_{u,n,t}^+$ or $a_{u,n,t}^-$ is activated, $x_{u,n,t}$ is forced to be 0 as per (13) and (14). Based on the objective function introduced in Section III-H, there is no need to consider the case where $x_{u,n,t}$ is forced to be 1, as it has a direct impact on the objective function and therefore its activation is guaranteed.

$$1 - x_{u,n,t} \geq a_{u,n,t}^+, \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (13)$$

$$1 - x_{u,n,t} \geq a_{u,n,t}^-, \quad \forall t \in \mathcal{T}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (14)$$

As it happened with the MCS selection variable, band assignment will occur only to users located in cells that have been chosen for illumination. Consequently, variables $b_{u,w,t}$ and $x_{u,n,t}$ are restricted by the illumination variable $i_{c,t}$ according to (15) and (16).

$$\sum_n x_{u,n,t} \leq \sum_c UpC(u, c) \cdot i_{c,t}, \quad \forall t \in \mathcal{T}, \forall u \in \mathcal{U} \quad (15)$$

$$\sum_w b_{u,w,t} \leq N \cdot \sum_c UpC(u, c) \cdot i_{c,t}, \quad \forall t \in \mathcal{T}, \forall u \in \mathcal{U} \quad (16)$$

D. Interference Management in Adjacent Cells

Adjacent cells are considered to interfere with each other if and only if they are illuminated simultaneously with the same band. As single color illumination is assumed, from the total B_T , the utilisation of same frequency bins for the simultaneous illumination of adjacent cells must be restricted. Adjacent cells are characterised by the precomputed **AdjC** matrix, and so are the users aggregated in those cells through the **AdjU**. That being so, the imposed restriction on frequency bin assignment is depicted in (17).

$$b_{u_\alpha, w, t} + b_{u_\beta, w, t} \leq 1, \quad \forall t \in \mathcal{T}, \quad \forall w \in \mathcal{N}, \alpha \neq \beta, \forall (\alpha, \beta) \in \mathbf{AdjU} \quad (17)$$

E. Operating Point Selection

The precomputation of possible offered capacity values as a function of illumination, power and bandwidth decision variables transforms the formulation into an integer linear programming formulation where just binary decisions must be taken by the BH algorithm. To ease the computation of the associated continuous discrete power and capacity values, and therefore the characterisation of timing and traffic counters, the inclusion of a new operating point variable is required: $z_{u,m,n,t}$. The operating point is defined as the band plus MCS configuration, and it therefore aggregates $m_{u,m,t}$ and $x_{u,n,t}$ related information. According to (18) $z_{u,m,n,t}$ is activated only if both $m_{u,m,t}$ and $x_{u,n,t}$ are activated. To force the activation of $m_{u,m,t}$ and $x_{u,n,t}$ when the optimizer takes a decision over $z_{u,m,n,t}$ (19) and (20) are required.

$$x_{u,n,t} + m_{u,m,t} - z_{u,m,n,t} \leq 1, \quad \forall t \in \mathcal{T}, \forall m \in \mathcal{M}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (18)$$

$$z_{u,m,n,t} \leq m_{u,m,t}, \quad \forall t \in \mathcal{T}, \forall m \in \mathcal{M}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (19)$$

$$z_{u,m,n,t} \leq x_{u,n,t}, \quad \forall t \in \mathcal{T}, \forall m \in \mathcal{M}, \forall n \in \mathcal{N}, \forall u \in \mathcal{U} \quad (20)$$

Through $z_{u,m,n,t}$ the required power can now be derived, as $P(u, m, n)$ is not only linked to the MCS selection $m \in \mathcal{M}$ but also to the assigned number of frequency bins $n \in \mathcal{N}$ in order to close the link budget of (1) and therefore offering capacity $D(m, n)$. At this point, the total RF transmit power restriction of (21) can be imposed as the sum of the power assigned to all users at each time instant can not exceed P_T .

$$\sum_u p_{u,t} = \sum_u \left(\sum_{m,n} P(u, m, n) \cdot z_{u,m,n,t} \right) \leq P_T, \quad \forall t \in \mathcal{T} \quad (21)$$

F. Traffic Counter

The requested capacity is already characterized for T_T through $R(u, t)$. The served capacity, $d_{u,t}$, depends on the operating point (m, n) selection, and so in $D(m, n)$.

$$d_{u,t} = \sum_{m,n} D(m, n) \cdot z_{u,m,n,t}, \quad \forall u \in \mathcal{U}, \forall t \in \mathcal{T} \quad (22)$$

From a traffic perspective, the algorithm aims to minimize the absolute value of the difference between these two parameters: $|d_{u,t} - R(u, t)|$. In order to linearize the expression, two additional variables are introduced, the EC, $f_{u,t}$, and the UC, $\gamma_{u,t}$. In (23) the extra-served capacity is accounted, as the positive difference between the served and requested capacity. In (24) the sign of the difference is swapped, and therefore the unserved capacity is considered. The requested capacity is accumulated over time, and thus, the remaining unserved capacity from the previous time instants, $\gamma_{u,t-1}$, must be factored in both expressions.

$$f_{u,t} \geq \left(\sum_{m,n} D(m, n) \cdot z_{u,m,n,t} \right) - R(u, t) - \gamma_{u,t-1}, \quad \forall t \in \mathcal{T}, \forall u \in \mathcal{U} \quad (23)$$

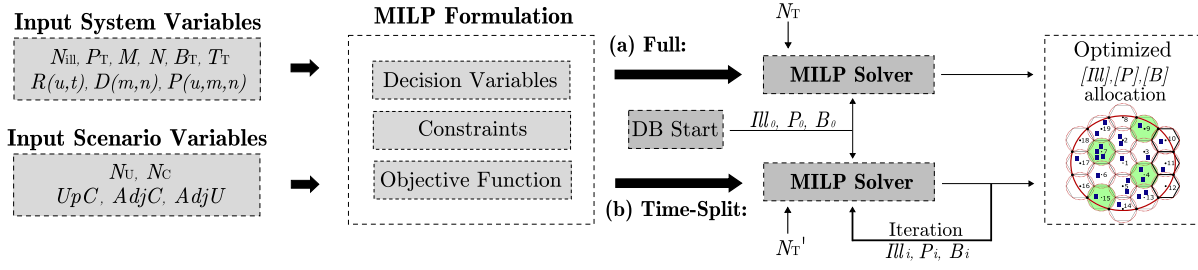


Fig. 4. MILP formulation to jointly optimize illumination, power and bandwidth. (a) Full MILP case, where the problem is solved for all N_{T} time instants. (b) Time-split MILP case, where the problem is iteratively solved for N_{T}' time instants at a time, with $N_{\text{T}}' \ll N_{\text{T}}$.

$$\gamma_{u,t} \geq R(u, t) + \gamma_{u,t-1} - \left(\sum_{m,n} D(m, n) \cdot z_{u,m,n,t} \right), \quad \forall t \in \mathcal{T}, \forall u \in \mathcal{U} \quad (24)$$

G. Timing Counter

In addition to the traffic concern, in the BH scheme users are not continuously illuminated and therefore cells are in competition to gain and retain radio resources. Consequently, the time-to-serve a given user must also be characterized. To this end, a TTS counter variable is introduced, $c_{u,t}$. As illumination is cell refereed, if the users within the cell keep demanding traffic or have pending traffic that has still not been served, the counter increases its value at each time instant. If the cell is illuminated at a given instant, the counter is reset to 0. The constraint is depicted in (25).

$$\frac{\gamma_{u,t}}{\sum_{c/u \in c} R(u, t)} + c_{u,t-1} - N_{\text{T}} \cdot i_{c,t} \leq c_{u,t}, \quad \forall t \in \mathcal{T}, \forall c \in \mathcal{C} \quad (25)$$

H. Objective Function

The problem is formulated as a single-objective optimization problem in (26). If different objectives coexist, these need to be weighted and integrated in a comparable manner. Together with the timing parameter, counting with two additional variables to represent independently excess and unmet capacity helps weighting separately each of the concepts and as a result allowing the operator to select the parameter that is aimed to be prioritized, based on service particularities. For that purpose, three weighting parameters are introduced, each of them multiplying the respective variable: w_{TTS} , w_{EC} and w_{UC} . To effectively combine objectives in the objective function, normalization is crucial as traffic and time-based variable values are inherently different and cannot be directly added: n_{TTS} , n_{EC} and n_{UC} . Normalizing these values ensures that weighting parameters accurately reflect their respective contributions, resulting in the desired representative effect in the fitness function. The election of weighing parameters and normalization factors is thoroughly described in Section IV-C.1.

$$\min \sum_{u,t} \left(\frac{w_{\text{UC}} \cdot \gamma_{u,t}}{n_{\text{UC}}} + \frac{w_{\text{EC}} \cdot f_{u,t}}{n_{\text{EC}}} \right) + \sum_{c,t} \frac{w_{\text{TTS}} \cdot c_{c,t}}{n_{\text{TTS}}} \quad (26)$$

s.t. (4), (5), (9) – (21), (23) – (25)

I. Computation Complexity

The complexity of the proposed formulation can be analyzed from a memory and an algorithmic perspective. Regarding the former, the memory requirement grows with the largest sizes of Table III: $i_{c,t}$ (or $c_{c,t}$) and $z_{u,m,n,t}$. These result in a memory complexity of $O(N_{\text{C}}N_{\text{T}} + N_{\text{U}}MN_{\text{T}})$. Regarding the latter, to assess the algorithmic complexity of the proposed formulation, we introduce parameter $N_{\text{U/C}}$, which represents the maximum number of users located in a single cell. $N_{\text{U/C}}$ is computed as $N_{\text{U/C}} = \max_c \sum_u U_{\text{pC}}(u, c)$. The total number of possible solutions scales with three factors: illumination ($i_{c,t}$), MCS selection ($m_{u,m,t}$), and band selection ($b_{u,w,t}$). Note that the auxiliary variables are fully defined once the previous variables are fixed. Given at most N_{ill} cells can be illuminated at time t , the number of possible illumination solutions for a given time scales with $O\left(\frac{N_{\text{C}}!}{(N_{\text{C}} - N_{\text{ill}})!}\right)$ (which $\lim_{N_{\text{C}}/N_{\text{ill}} \rightarrow \infty} = N_{\text{C}}^{N_{\text{ill}}}$). At each time, each sub-band can only be assigned to one user, out of, at most, $N_{\text{U/C}}N_{\text{ill}}$ users. Since there are N_{ill} subbands due to the frequency reuse, this leads to $O((N_{\text{U/C}}N_{\text{ill}})^{N_{\text{ill}}})$. Similarly, at each time, each user can choose among M MCSs, leading to $O(M^{N_{\text{U/C}}N_{\text{ill}}})$. The total possible number of solutions is then $O(N_{\text{C}}^{N_{\text{ill}}N_{\text{T}}}(N_{\text{U/C}}N_{\text{ill}})^{N_{\text{ill}}N_{\text{T}}}M^{N_{\text{U/C}}N_{\text{ill}}N_{\text{T}}})$. For a classical mathematical solver, as the one utilized in this work, this is also the algorithmic complexity.

J. Time-Split Resolution

As shown in the previous paragraph, the algorithmic complexity of MILP is driven partially by N_{T} . To reduce the dimensionality of the problem, then, one effective strategy is to bound N_{T} by dividing the time horizon into $\frac{N_{\text{T}}}{N_{\text{T}}'}$ disjoint subspaces, each containing N_{T}' time slots. Each subproblem is then independently solved over its respective subset.

$$\mathcal{T} = \mathcal{T}_1 \cup \mathcal{T}_2 \cup \dots \cup \mathcal{T}_{\frac{N_{\text{T}}}{N_{\text{T}}'}} \quad (27)$$

Splitting the time into smaller subspaces reduces the size of N_{T} to the size of the subspace under consideration, thereby bounding the computational complexity discussed above. For each subproblem, the computational complexity becomes $O(N_{\text{C}}^{N_{\text{ill}}N_{\text{T}}'}(N_{\text{U/C}}N_{\text{ill}})^{N_{\text{ill}}N_{\text{T}}'}M^{N_{\text{U/C}}N_{\text{ill}}N_{\text{T}}'})$.

Since there are $\frac{N_{\text{T}}}{N_{\text{T}}'}$ such subproblems, the total computational complexity of the entire approach is $O\left(\frac{N_{\text{T}}}{N_{\text{T}}'}N_{\text{C}}^{N_{\text{ill}}N_{\text{T}}'}(N_{\text{U/C}}N_{\text{ill}})^{N_{\text{ill}}N_{\text{T}}'}M^{N_{\text{U/C}}N_{\text{ill}}N_{\text{T}}'}\right)$.

This decomposition leads to a significant computational gain, as the exponential dependence shifts from N_T to N_T' . Provided that $N_T' \ll N_T$, this approach results in a substantial reduction in overall computational cost, making the solution more scalable.

As depicted in Fig. 4, and further developed in Algorithm 1, the MILP solver is iteratively executed. From iteration to iteration, only the decision variables with time indexes contained within $\mathcal{T}_i$ are activated. Variables with past time indexes are fixed based on the output values obtained from previous evaluations. In terms of constraints, only the ones affecting the ongoing evaluation are considered.

The resolution of the full MILP is unfeasible in systems of high-dimensionality linked to a high number of users or time. Therefore transitioning into a time-split MILP allows the inclusion of such a formulation, that, as it will be seen does not significantly degrade the performance of the system.

The source code corresponding to full and time-split MILP algorithms is available on [32].

IV. SIMULATION RESULTS

This section outlines the simulation environment used to evaluate the satellite system under study, including user distribution and simulation scenarios. It details the simulation setup, highlighting the MILP solver and baseline deterministic DB and GA methods used for comparison. The section also discusses how weighting and convergence parameters affect UC, EC and TTS performance metrics, allowing satellite operators to select these flexibly. A convergence study helps identify the optimal objective gap threshold. Results compare MILP and time-split MILP in both small- and large-scale scenarios, demonstrating scalability and effectiveness.

A. Simulation Environment

A satellite in LEO orbit at an altitude of h_{sat} is considered. The minimum elevation angle from user terminals is el_{min} . The antenna from [10] is used as a reference, determining for each h_{sat} and el_{min} pair, the beamwidth ($\Delta\theta_{3\text{dB}}$) associated to a total N_C fixed-cell scheme. The antenna complexity is fixed and considered invariant, thus, same beamwidth cells are generated.

With the aim of optimizing as much as possible the existing spectrum, together with the timing diversity exploited through the beam-hopping strategy, the spatial diversity is also intended to be utilized, counting with a high cell number distribution: N_C . The number of cells within the scenario directly conditions the required frequency diversity. Scenarios with a large number of cells allow, under the restriction of not illuminating adjacent cells with the same band, decreasing the number of required frequency bins (N) within B_T . However, as the number of cells within the field of view increases, the required beamwidth narrows, resulting in higher directivity and, consequently, a larger antenna aperture.

The maximum transmit power value (P_T) is determined for a 12U small satellite platform [10] in LEO orbit, a feasible value for the number of users (N_U) that are located within the satellite field of view. The available MCS set, M , is depicted

Algorithm 1 MILP-Based Beam-Hopping With Full and Time-Split Execution

Require: $h_{\text{sat}}, el_{\text{min}}, f_c, N_T, T_T, N_{\text{ill}}, N, M, N_U, D_U, N_C, P_T, B_T$

Ensure: Illumination (**Ill**), Bandwidth (**B**), Power (**P**); KPIs {UC, EC, TTS}

- 1: **Pre-computation:**
- 2: Distribute N_U users within the FoV of the satellite following distribution model
- 3: Generate N_C ring-based cell layout and compute $UpC(\cdot)$, $AdjC(\cdot)$, $AdjU(\cdot)$
- 4: Build lookup variables $D(\cdot)$, $P(\cdot)$ and traffic $R(\cdot)$ over $\mathcal{T}$ as per D_U
- 5: **Demand-based start:**
- 6: Run DB BH $\rightarrow (\text{Ill}_0, \mathbf{B}_0, \mathbf{P}_0)$; convert to MIP start i_0, b_0, x_0, m_0, z_0
- 7: **MILP definition:**
- 8: Declare decision variables $i, m, b, z, x, a^+, a^-, \gamma, f, c$
- 9: Add constraints (4), (5), (9)–(21), (23)–(25)
- 10: Define single objective function (26) with weights $(\beta^*, (1 - \beta^*)/2, (1 - \beta^*)/2)$ for {UC, EC, TTS}
- 11: **Time partition setup:**
- 12: **if** method = *Full-MILP* **then**
- 13: $\mathcal{P} \leftarrow \{\mathcal{T}\}$
- 14: **else if** method = *Time-Split-MILP* **then**
- 15: Partition $\mathcal{T}$ into $\mathcal{P} = \{\mathcal{T}_i\}_{i=1}^{N_T/N_T'}$ with $|\mathcal{T}_i| = N_T'$
- 16: **end if**
- 17: **for** $i \leftarrow 1$ to $|\mathcal{P}|$ **do**
- 18: Fix all decision variables with time indices in $\bigcup_{j < i} \mathcal{T}_j$ to solved values
- 19: Restrict all time-indexed constraints (4), (5), (9)–(21), (23)–(25) to indices $t \in \mathcal{T}_i$
- 20: **MILP Resolution:**
- 21: **for** $\beta \in \{0, 1, \beta^*\}$ **do**
- 22: Set MIPGap: coarse for $\beta \in \{0, 1\}$, tight for β^* .
- 23: When $\beta = \beta^*$, set normalization: $n_{\text{UC}} \leftarrow \text{UC}_{\text{max}}$ ($\beta = 0$) and $n_{\text{EC}}, n_{\text{TTS}} \leftarrow \text{EC}_{\text{max}}, \text{TTS}_{\text{max}}$ ($\beta = 1$)
- 24: Load DB MIP start: DB MIP start projected to $\mathcal{T}_i$ and previous-partition MILP solution: i_i, b_i, x_i, m_i, z_i
- 25: Apply time-window bounds to decision variables via (lb, ub)
- 26: Launch MILP solver on $\mathcal{T}_i$
- 27: Extract and append partition solution to global: (**Ill** _{i}), (**B** _{i}) and (**P** _{i})
- 28: **end for**
- 29: Persist solved values for $t \in \mathcal{T}_i$ by tightening (lb, ub)
- 30: **end for**
- 31: For final appended solution (**Ill** = **Ill** _{$|\mathcal{P}|$} , **B** = **B** _{$|\mathcal{P}|$} and **P** = **P** _{$|\mathcal{P}|$})
- 32: Compute final KPIs over $t \in \mathcal{T}$ from the assembled (**Ill**, **B**, **P**)

in Fig. 3 and according to (1), the point in which each user will operate directly depends on the location of each user and the specifications of the associated user terminal. The user

TABLE IV
INPUT SCENARIO

Variable	Reduced	Enlarged
h_{sat}	1100 km (LEO)	
f_c	20 GHz (Ka)	
e_{min}	25°	
$A_{\text{footprint}}$	8e6 km ²	
$\Delta\theta_{\text{3dB}}$	32.20°	11.85°
N_{ill}	4 beams	
T_{T}	3 s	10 s
N_{T}	30 time slots	100 time slots
SCS	15 kHz	
N_{C}	37 cells	271 cells
N_{U}	[5,30] users	[5,100] users
D_{U}	U(0,30) Mbps	
P_{T}	18 W	
M	15	
B_{T}	125 MHz	
N	5	
α	10 %	

distribution is randomly generated among the satellite field of view, with an associated demand per user of D_{U} .

As described, due to inherent LEO satellite propagation particularities with respect to Earth's surface, the total simulation time is defined to be able to consider a fixed screenshot during T_{T} . Among the available numerology types in 5G NR, the 15 kHz sub-carrier spacing (SCS), which coexists with LTE, is selected as a representative use case for this study. The illumination granularity is set to $T_{\text{T}}/N_{\text{T}}$ to balance KPI accuracy, computational complexity of the illumination sequence, and the technical feasibility of beam switching. A higher BH granularity, thus a lower $T_{\text{T}}/N_{\text{T}}$ value, enhances traffic and service time-based performance. Conversely, lower granularity, improves beam stability, reduces power consumption, and simplifies network signaling, albeit at the cost of reduced flexibility.

Simulation experiments have been divided to assess two key aspects: first, a comparison of different optimizers, including MILP and MILP time-split, is conducted in a low-dimensional scenario with a limited number of users and time slots (reduced scenario). Second, the performance of the optimizers is analyzed in a scenario with a larger number of users and extended simulation time (enlarged scenario). The separation between reduced and enlarged scenarios is driven by the exponential growth in complexity of the MILP formulation as elaborated in Section III-I. In Table IV the two reference scenarios for performance evaluation are presented.

In total 10 different user distribution and demand scenarios are simulated, by averaging the obtained KPIs for each reference scenario: reduced scenario and enlarged scenario.

B. Simulation Setup

For the experimental scenarios presented in this section, the analytical MILP results are compared with the instantaneous DB methodology introduced in [10], as well as a multi-objective NSGA-II-based approach proposed by the authors in [33], referred to as GA throughout the paper. Within the GA approach, mutation and crossover operators are defined, modifying from iteration to iteration illumination, power and bandwidth allocation. Additionally, following the

TABLE V
GA CONFIGURATION PARAMETERS

Parameter	Value
Num. of Generations	100 K
Population Size	100
Crossover Rate	1.0
Mutation Rate	0.1
DB Local Search Rate	0.2

TABLE VI
GUROBI CONFIGURATION PARAMETERS

Parameter	Full	Split
MIPGap	20 %	1 %
β	0.7	
Threads	4	
Method	Barrier	
Heuristics	0.05	
Presolve	Active	

demand-based logic, a DB based local search operator is designed, to help accelerating the convergence of the meta-heuristic solver. The GA configuration parameter values and operator rates are summarized in Table V. The output of the multi-objective evolutionary algorithm corresponds to a set of solutions that includes all non-dominated objective values for the problem. It is important to note that, for comparison purposes, the selection of a solution from the Pareto front is carried out based on a balanced criterion among the UC, EC, and TTS parameters.

The mathematical solver employed for solving the presented MILP problem is Gurobi (version 11.0.0), a robust commercial solver suitable for solving large-scale optimization models such as the presented high-dimensional MILP problem. For the sake of balancing the three parameters of interest, two traffic-based (UC and EC) and the time-based one (TTS), the tunable weighting hyperparameter β is introduced, described in Section IV-C.1. The configuration parameters used by the solver are summarized in Table VI.

C. Hyperparameter Selection

This section examines the selection of key parameters that are not strictly defined by the scenario but arise from the problem formulation. First, it discusses weighting and normalization, which let operators prioritize objectives. Then, it covers convergence evaluation, helping balance solution accuracy and computational efficiency. The aim is to guide effective parameter selection for the ongoing optimization process.

1) *Weighting Factors and Normalization Parameter Study:* Within illumination and resource allocation problems, the unserved capacity, UC, is considered the key parameter of interest as it characterizes the system performance. However, to evaluate the effect on EC and TTS we consider a weighted single-objective minimization jointly considering UC, EC and TTS. UC quantifies the benefit of the function, in terms of throughput, while TTS and EC quantify the cost, in terms of quality of service and usage of resources. To model appropriate weighting, the tunable hyperparameter β is introduced, transforming (26) into (28). Since both EC and TTS quantify distinct aspects of cost, these are combined under the same

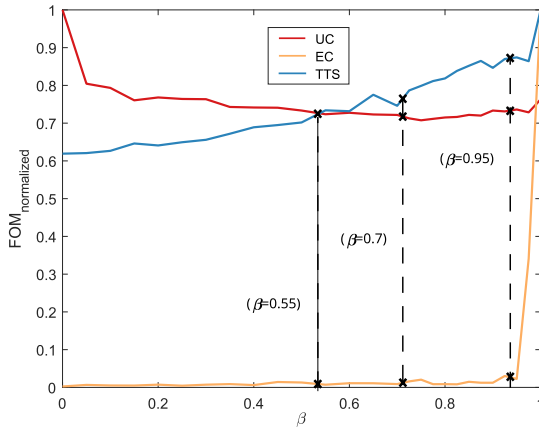


Fig. 5. Weighting parameter evaluation.

weighting parameter $(1 - \beta)$. The value of β defines the operating point within the UC-EC-TTS solution space and can be adjusted according to the specific requirements of the service to be provided. High values of β prioritize the reduction of UC, making them suitable for scenarios with high-capacity service demands. However, this focus leads to increased EC and longer time-to-serve TTS. Conversely, when the objective is to reduce power waste or to meet strict latency requirements, lower values of β are preferred, as they shift the trade-off toward minimizing EC and TTS.

$$\min \sum_{u,t} \frac{\beta \cdot \gamma_{u,t}}{n_{UC}} + \frac{(1-\beta)/2 \cdot f_{u,t}}{n_{EC}} + \sum_{c,t} \frac{(1-\beta)/2 \cdot c_{c,t}}{n_{TTS}} \quad (28)$$

s.t. (4), (5), (9) – (21), (23) – (25)

Additionally, within (28), n_{UC} , n_{EC} and n_{TTS} values need to be determined. Since the value ranges and measurement units of UC, EC, and TTS differ significantly, it is crucial to normalize these parameters within the objective function to ensure a fair weighting. The normalization is scenery dependent, meaning that for each user demand and distribution, the normalization parameters are computed first. The presented MILP problem is solved twice for that purpose. For determining n_{UC} , (28) is solved fixing $\beta = 0$, and thus minimizing EC and TTS. For n_{EC} and n_{TTS} , $\beta = 1$, by just minimizing UC. Once the normalization parameters are determined and the weighting parameter is fixed based on the particularities of the service of interest, the presented problem can now be solved, either through the full or time-split optimization strategies presented below.

In Fig. 5, the relationship between the three parameters is represented as a function of the weighting parameter. As it is expected, for $\beta = 1$, both EC and TTS take the maximum value, whereas for $\beta = 0$, it is UC the one that is maximized. For β values that get away from the unity, it can be seen how the EC is drastically reduced, being almost insignificant in its behavior with β under the $\beta < 0.95$ condition. The cross point between UC and TTS occurs at around $\beta = 0.55$, where a compromise exists between unserved capacity and the time-to-serve.

For the three weighting configurations marked in Fig. 5, the UC, EC, and TTS parameters are evaluated in a 50-user scenario using the time-split MILP methodology. The influence of

TABLE VII
ENLARGED SCENARIO RESULTS AS A FUNCTION OF β

	Weighting Parameter		
	$\beta = 0.55$	$\beta = 0.7$	$\beta = 0.95$
UC [%]	10.66	9.81	9.68
EC [%]	18.64	20.77	23.63
TTS [s]	1.40	2.26	2.93

TABLE VIII
SMALL-SCALE USER SCENARIO CONVERGENCE
EVALUATION: COMPUTATIONAL TIME

MIP Gap	Number of users		
	5u	6u	7u
20%	0.003h	0.0038h	0.0048h
1%	2h	10h	80h
0.001%	7h	>100h	>100h

β on these three parameters is intended to be quantized through the presented example. Table VII summarizes the obtained KPIs.

The TTS parameter exhibits a significantly steeper slope compared to UC and EC, demonstrating a higher sensitivity to the weighting factor. In contrast, within the given β range, both UC and EC remain relatively stable. For the ongoing simulations, the service-dependent weighting parameter is set to $\beta = 0.7$ as a balanced trade-off among the considered parameters. Ultimately, the choice of β depends on the priorities of the satellite operator, allowing for customization based on specific service requirements.

2) *Convergence Evaluation:* The MIPGap parameter defines the permissible difference between the best-known feasible solution and the best-known bound on the optimal solution. This upper bound ratio indicates the proximity of the current solution to optimality. Given the high-dimensional nature of the problem under study, achieving a zero gap is computationally intractable. To determine the most suitable MIPGap, the convergence behavior of different problem instances with increasing complexity. Specifically, we examined three scenarios: one with 5 users and observing the time to converge at 0.001%, 1%, and 20%, and two with 6 and 7 users, respectively, observing the time to converge at 1% and 20%. Based on these results, the True Gap is computed, as the percentage difference between the best solution obtained by the solver and the overall optimal solution, which is approximated using the results from the 0.001% case for 5 users, and the 1% case for 6 and 7 users.

For small-scale scenarios (5–7 users), the results show that obtaining a solution with a 1% MIPGap is computationally feasible within hours when using the full MILP methodology. Furthermore, the difference between the solutions obtained with a 1% MIPGap and those obtained with a 0.001% MIPGap is minimal, indicating that a 1% threshold is sufficient for ensuring high-quality solutions. Importantly, the time required for convergence increases exponentially as the number of users grows, even when using a 1% MIPGap, as shown in Table VIII.

It is also important to note that while the MIPGap defines a stopping criterion, it does not directly correspond to the actual optimality gap. Empirical results suggest that the true gap is often significantly lower than the defined MIPGap, as

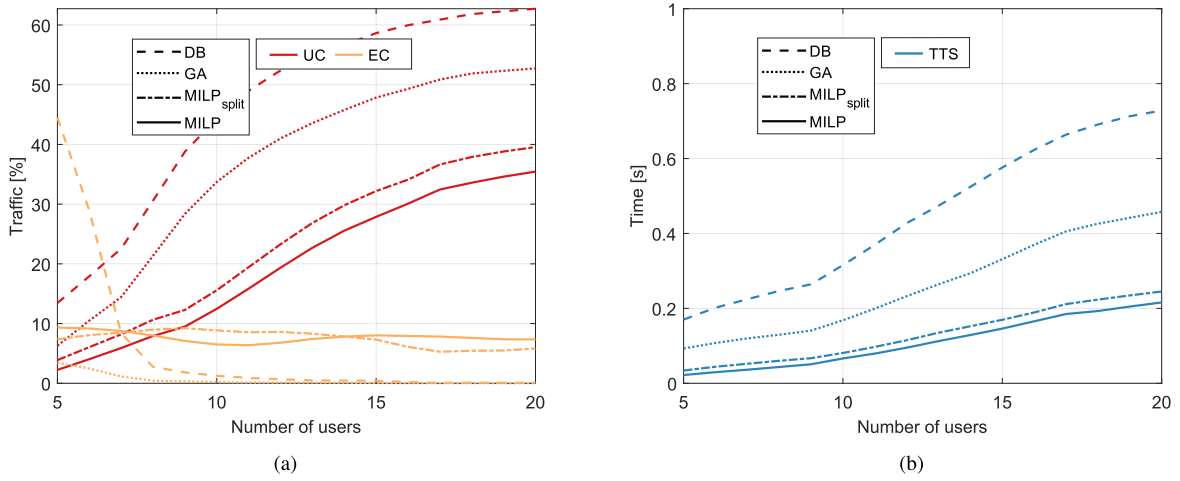


Fig. 6. Reduced scenario comparison: DB, GA, MILP, MILPsplit. (a) Traffic-based PoIs: UC and EC. (b) Time-based PoI: TTS.

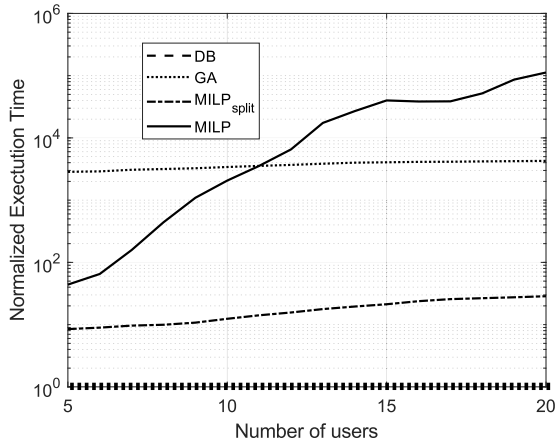


Fig. 7. Reduced scenario, computational time comparison: DB, GA, MILP, MILPsplit.

TABLE IX
TRUE GAP FOR 20% MIP GAP

	Number of users		
	5u	6u	7u
True Gap (%)	8	11.9	16.5

illustrated in Table IX. This implies that solutions obtained with a 1% MIPGap provide a very close approximation to the true optimal solution, while even a 20% MIPGap often results in solutions with a much lower true gap.

Within the time-split MILP methodology, increasing the number of users (N_U) does not impose such a significant computational burden, as relaxation exists within the time dimension (N_T). Consequently, maintaining a 1% MIPGap remains a practical and efficient choice even for larger-scale scenarios. These findings demonstrate that using a 1% MIPGap balances computational efficiency with solution accuracy, while a 20% MIPGap provides an acceptable approximation for high-dimensional problems where reducing the computational cost is a priority.

D. Performance Evaluation

For the reduced scenario, presented in Table IV, an extensive comparison between the optimal full MILP and the suboptimal

time-split MILP approaches is presented first. In addition, the normalized execution times are compared.

Once the benefits of the time-split MILP methodology are highlighted within the reduced scenario, with promising results in terms of KPIs and significant improvements in execution time, the enlarged scenario is then addressed using the time-split resolution methodology.

1) *Reduced Scenario*: The objective in the reduced scenario is to compare the two analytical exact optimization approaches from Fig. 4: full MILP and time-split MILP. In the former, 30 time slots are jointly optimized, while in the latter, the optimization is performed iteratively and cumulatively over periods of 10 time slots. For that purpose, a simplified scenario is presented, where BH is analyzed for a shorter time period as per Table IV. Scenarios of 37 cells are simulated while varying the number of users within the scenario from 5 to 20, therefore progressively increasing the requested capacity.

In Fig. 6a, UC and EC are evaluated. Both traffic-based parameters are presented as a percentage of the requested capacity that increases with the number of users. The GA method improves the deterministic DB method from [10], and as expected, the presented MILP formulation outperforms both. For UC, the full MILP optimization methodology represents the optimum solution, being able to evaluate the gap with respect to other joint illumination, power and bandwidth allocation solutions. For the defined GA configuration, the DB is surpassed on average by almost 10%. The time-split MILP already achieves a gain of 22% with respect to DB and the full MILP baseline an average gain of 25.5%. As it can be seen, the trend for both MILP approaches is quite similar, having identified computationally speaking, a simpler time-split MILP methodology that gets closer to the optimal full MILP. In terms of EC, for the selected weighting parameter, on average full and time-split MILP provide almost identical results, really effective for low user number schemes, where resources are underused through the DB methodology.

In terms of timing, the TTS results presented in Fig. 6b lead to similar conclusions. The GA achieves nearly twice the improvement over the DB method, while the MILP time-split methodology closely approaches full MILP performance,

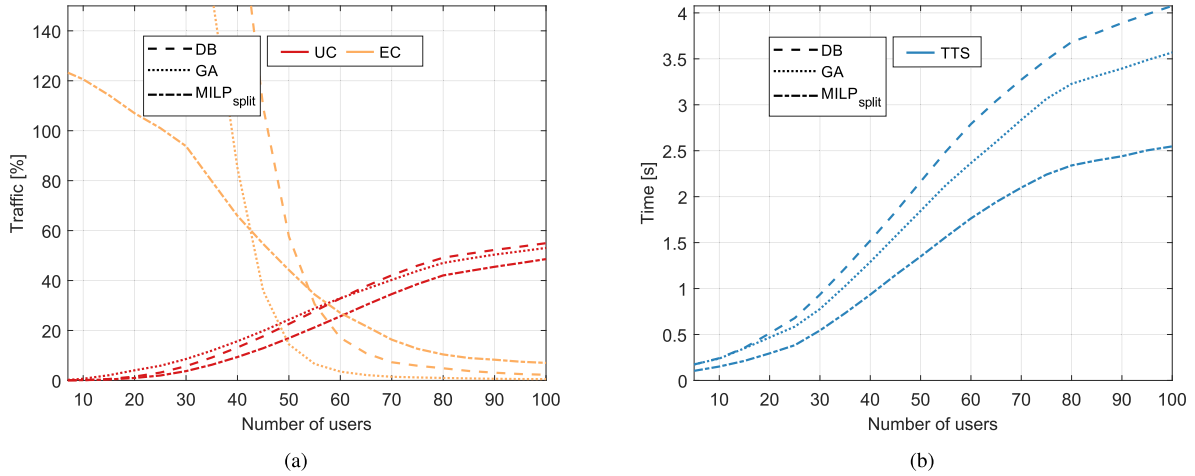


Fig. 8. Enlarged scenario comparison: DB, GA, MILPsplit. (a) Traffic-based PoIs: UC and EC. (b) Time-based PoI: TTS.

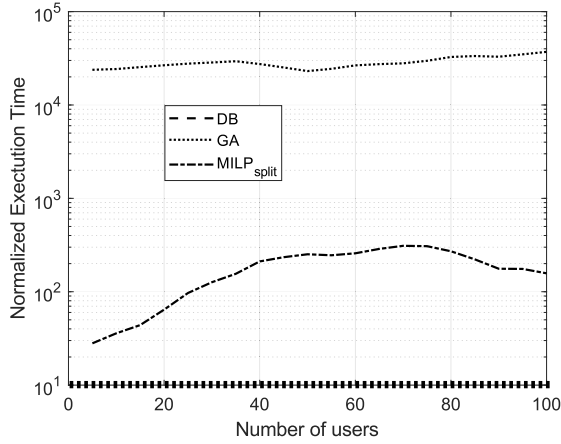


Fig. 9. Enlarged scenario, computational time comparison: DB, GA, MILPsplit.

achieving a gain of approximately five times compared to DB for any number of users. A fivefold improvement in TTS means that users will, on average, receive illumination five times faster with the MILP approach compared to the deterministic DB method. This results in a significant reduction in waiting time, enhancing responsiveness and efficiency in resource allocation. The results obtained with the time-split MILP remain nearly identical to those of the full MILP, with a slight advantage from the full MILP method.

Through both figures, the benefits of the exact time-split resolution method are clearly highlighted at both the traffic and timing levels. The close alignment of the MILP and time-split MILP results demonstrates that the time-split approach effectively approximates the optimal solution, preserving solution quality while providing a practical and efficient alternative for larger-scale problems.

In Fig. 7, normalized computational times are presented with respect to the instantaneous DB resolution method. GA supposes a two-order difference for the acquisition of the result and the full MILP, a difference from 2 to 5 orders of magnitude, depending on the number of users to be evaluated within the [5,20] user range. The greater the number of users, the higher the dimensionality of the problem, and consequently

the required computational time increases significantly. It is the time-split MILP method, the one that at a computational level clearly outperforms the full MILP method but also the GA approach, with better UC, EC and TTS parameters.

2) *Enlarged Scenario*: Building on the benefits of the time-split MILP methodology outlined in the previous section and taking into account the computational challenges of solving the full MILP model in a larger scenario with greater time and user dimensions, this section provides a comparison of the time-split MILP approach with the DB and GA methods for the enlarged scenario. In particular, according to Table IV, scenarios of 271 cells are evaluated, considering up to 100 users within it. Illumination, band and power allocation solutions are iteratively and cumulatively provided for periods of 10 time slots, out to a total of 100 time slots, each counting with a corresponding unique resource allocation solution that jointly minimizes the objective function.

As depicted in Fig. 8a, the MILP time-split methodology outperforms the UC of the DB method by an average of 5% and the GA solution by 6%, across the entire number of users. Regarding EC, two different behaviors are observed. For low number of users (up to 40), the MILP reduces EC with respect to DB and GA methods, proving to obtain better results on all three metrics. As the number of users increases, for the chosen weighting parameter β the MILP solution sacrifices the EC metric to obtain better UC and TTS. In other words, to obtain a better solution, the solver allows for a higher margin on extra resources to reduce the number of non-served users, as well as the time to serve them. In this line, for high number of users, the MILP solution achieves higher EC values than the DB and GA. At traffic level, an improvement of 6.5% for 100 users implies, for instance, a reduction on average of almost 100Mbps of unserved capacity.

The results for the TTS is presented in Fig. 8b. On average, MILP solutions within the user sweep achieve nearly twice the performance of the DB BH methodology. While GA enhances DB solutions significantly, they still remain approximately 1.5 times lower than MILP performance. Within the 10s (T_T) simulation, for 100 users (N_U) and just 4 active simultaneous beams (N_{III}), users need to normally wait around 2.5s through

MILP, opposed to almost 4s when employing DB and 3.5s when compared to the metaheuristic GA-based solution. The combination of a high user density, fine cell discretization, and extended time slot duration explains the overall increase in absolute TTS values compared to the reduced scenario. Reducing the time slot duration would, in turn, decrease the absolute TTS.

The computational time results in Fig. 9 follow a trend similar to that discussed for the reduced scenario. The GA execution time is controlled by the number of generations defined in the GA configuration values of Table V, showing an almost constant behavior for the entire user sweep. The normalized execution time increases by an order of magnitude in the enlarged scenario compared to the reduced one, precisely due to the increase of N_T .

Similarly, when transitioning to the enlarged scenario, the computational time for the time-split MILP methodology follows a similar trend, remaining almost constant for the entire user sweep, albeit it being about one order of magnitude higher. Above 40 users, no noticeable increase is appreciated as happened for the full MILP method in Fig. 7. That is due to the benefit of the time-split method, which allows a controlled execution time, clearly outperforming the selected GA method, both at parameter result and computational time.

V. CONCLUSION

This paper presents a joint optimization of power, bandwidth, and illumination time using beam-hopping payloads in NGSO constellations for broadband services. While existing research addresses BH challenges, most are focused on isolated aspects, often fixing frequency allocation or illumination patterns, or ignoring key indicators like user fairness, or excess of allocated resources. We propose an original analytical MILP formulation that jointly optimizes illumination sequence, bandwidth, and power allocation, subject to on-board antenna architecture, interference, and available power and spectrum resources.

The proposed objective function, in addition to minimizing unserved capacity, also reduces unnecessary extra resource allocation, critical in power-constrained systems, using the extra-served capacity parameter. Fairness in service delivery to users is further ensured through the time-to-serve parameter. The three parameters are integrated into a single objective optimization function, allowing the system operator the flexibility to select an appropriate operating point as needed.

To address the high computational complexity of the full MILP, a time-split MILP approach is presented and validated, dividing the problem over time into sub-problems that are sequentially solved. This method preserves the analytical structure of the MILP while aligning better with the temporal evolution of LEO satellite systems, where satellite movement continuously brings new users and cells into coverage. Unlike static, full-horizon methods, the time-split MILP formulation supports incremental decision-making and more accurately reflects the changing optimization landscape over time.

The study reveals that the time-split MILP methodology significantly outperforms the demand-based and genetic algorithm methods in terms of UC, EC, and TTS. For the

reduced scenario, the time-split MILP method shows a notable improvement of 22% over DB in UC, matching the performance of the full MILP baseline. For EC, both full and time-split MILP methods achieve similar, optimal results, while TTS performance is nearly doubled by MILP compared to DB and surpasses GA results. In the enlarged scenario, the superiority of the time-split MILP is demonstrated. For a 10-second simulation with 100 users and 4 active beams, almost 100Mbps of unserved capacity is reduced on average, corresponding to an improvement of 6.5%, with users waiting 2.5 seconds to be served with MILP, compared to nearly 4.5 seconds with DB and 3.5 seconds with GA. The time-split method maintains computational feasibility within the same order of magnitude across the tested user scenarios, highlighting its strength in balancing execution time, surpassing GA and the full MILP approach, and additionally in performance, surpassing DB and GA in result quality both for the two traffic-based and the time-based parameters.

In addition to the DB and GA solutions introduced by the authors for comparison, the current formulation could serve as a reference for assessing other BH optimization methods, including deterministic, heuristic, meta-heuristic, and artificial intelligence approaches. The proposed MILP framework provides a reliable reference for evaluating the effectiveness of these optimization strategies in satellite resource allocation, enabling the quantification of optimality gaps and offering crucial insights into the distance of individual solutions from the theoretical optimum. This capability not only enhances the assessment of current methods but also lays the groundwork for the development of data-driven approaches. As a future research direction, the insights from this study could guide the training of resource allocation models using DRL, facilitating further progress in resource allocation optimization.

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